

Psych 4110 Exam 1: Textbook Review Questions and Answers

Every end-of-chapter "Review Exam" question from textbook Chapters 1–5, with answers, plus the Chapter 5 drug classification table.

Answers marked **(book)** cover material that wasn't in the lectures or slides you have. They come from standard textbook content, so check them against your book if you can.

Chapter 1: Basic Pharmacology (how drugs enter, navigate through, and exit the body)

1. **What is the definition of a "drug"?** A chemical substance that, once in the body, alters some normal physiological function. Every medicine is a drug; "drug" vs. "medicine" is about connotation and context, not chemistry.
2. **What is the general flowchart of events when a drug is taken?** Administration → absorption into the blood → distribution to compartments (crossing the BBB for brain drugs) → binding at receptors (the drug's effects) → biotransformation by the liver → excretion by the kidneys.
3. **Difference between "enteral" and "parenteral" routes?** Enteral goes through the alimentary canal (GI tract), e.g., swallowing a pill. Parenteral is every route that bypasses the GI tract.
4. **Major limitations of the enteral route? (book)** Slow and unpredictable absorption (about 20–30 minutes, varies with food in the stomach, as with edibles); stomach acid and digestive enzymes can destroy some drugs; large molecules like peptides get broken down; the drug passes through the liver before reaching general circulation (first-pass metabolism); dose is hard to control.
5. **Name the three classes of parenteral routes.** **Injection** (IV, IM, subcutaneous), **pulmonary** (inhalation), and **topical** (sublingual, intranasal, on the skin).
6. **Significance of a drug entering circulation?** Once in the blood, the drug reaches every compartment of the body, including the brain. That's why the bloodstream is the key step.
7. **Structural components of the cell membrane?** A phospholipid bilayer (a double layer of fat) with embedded proteins, such as receptors, channels and transporters. Drugs must be lipid soluble to cross it unless a specific transporter exists.
8. **Define "absorption."** The time it takes for a population of drug to enter the blood after administration.
9. **Difference between "bound" and "free" drug molecules? (book)** Bound molecules are attached to blood proteins like albumin; they can't leave the blood or act on receptors, so they act as a reservoir. Free molecules are unattached; only free drug can diffuse out of capillaries and produce effects.
10. **What is "lipid solubility"?** A molecule's ability to dissolve in fat. It lets drugs cross cell membranes and the blood-brain barrier, and makes them store in fat (long half-life) and resist excretion in urine.
11. **What is the blood-brain barrier and how does it affect how drugs influence behavior?** Brain capillaries with tight gaps plus surrounding glial cells. A molecule must cross 4 membranes to reach brain tissue. Only lipid-soluble drugs get in easily, so the BBB decides which drugs can change behavior. It exists because neurons are lost daily and not replaced, so toxins must be kept out.

12. **Two steps in getting a drug out of the body?** (1) Biotransformation by the liver into a less lipid-soluble form; (2) filtering from the blood into urine by the kidneys.
13. **Name of the enzyme family that biotransforms most drugs?** Cytochrome P450 (CYP450). Six of these enzymes handle 90% of drugs. Metabolizer types: poor ~10%, intermediate ~10%, extensive ~78%, ultra-rapid ~2%.
14. **What is "half-life" and why does it matter?** The time to clear half of an initial dose from the blood. About 4 half-lives clear 94%. It tells you how long a drug keeps acting and how much is left when the next dose is taken; slow clearance means accumulation and possible toxicity.
15. **Name the different types of "drug receptors."** Receptors for neurotransmitters, enzymes, and membrane transport mechanisms (like reuptake transporters).
16. **Ways to express dose-response curves?** (1) Effect intensity vs. dose (rises, then plateaus); (2) frequency distribution: how many subjects respond at each dose; (3) cumulative percent of subjects showing the effect (best for reading ED values).
17. **What is the ED50?** The dose at which 50% of the population shows the effect.
18. **What is the LD50?** The dose that kills 50% of subjects (determined in animal research).
19. **What is the therapeutic index (TI)?** $LD_{50} \div ED_{50}$. Bigger is safer. Book example: an ADHD drug with $LD_{50} = 130$ has a TI of $130/30 \approx 4.3$ for improved attention, but only $130/70 \approx 1.86$ for heart palpitations. The palpitation curve sits "closer" to the lethal curve, so the attention effect is the safer one.

Chapter 2: The nervous system, neurons and neurotransmitters

1. **Difference between the CNS and the PNS?** The central nervous system is the brain and spinal cord. The peripheral nervous system is all the nerves outside them (somatic and autonomic).
2. **Two divisions of the autonomic nervous system? (book)** Sympathetic and parasympathetic.
3. **What does the sympathetic division do? (book)** "Fight or flight": raises heart rate and blood pressure, dilates pupils, releases glucose, slows digestion. (The parasympathetic does "rest and digest.")
4. **Five basic neuroanatomical divisions of the brain?** Telencephalon, diencephalon, mesencephalon, metencephalon, myelencephalon.
5. **Name a structure for each division. (book)** - Telencephalon: cerebral cortex (also basal ganglia, hippocampus, amygdala) - Diencephalon: thalamus, hypothalamus - Mesencephalon (midbrain): substantia nigra, ventral tegmental area (VTA) - Metencephalon: pons, cerebellum - Myelencephalon: medulla
6. **What is the limbic system?** A collection of interconnected nuclei that regulates emotion (not strictly one circuit).
7. **Describe one specific functional "system."** Pick one: **nigrostriatal** (substantia nigra → basal ganglia; coordinates behavior and choices; its loss causes Parkinson's), **medial forebrain bundle** (arousal and reinforcement), or **VTA → nucleus accumbens** (craving, liking and wanting drugs; part of a larger limbic circuit with frontal cortex, hippocampus, amygdala).
8. **Components of the synapse? (book)** The presynaptic terminal (with vesicles of neurotransmitter), the synaptic cleft, and the postsynaptic membrane with receptors. Some synapses also have presynaptic receptors.

9. **What is a "vesicle"? (book)** A small membrane sac in the presynaptic terminal that stores neurotransmitter.
10. **How does "exocytosis" occur? (book)** An action potential reaches the terminal, calcium flows in, vesicles fuse with the presynaptic membrane, and they release neurotransmitter into the cleft.
11. **Ways a neuron recovers from exocytosis? (book)** Reuptake of neurotransmitter by transporters (e.g., serotonin and dopamine reuptake transporters), enzymatic breakdown in the cleft (e.g., MAO for monoamines), diffusion away, and recycling the vesicle membrane back into the terminal.
12. **Classes of neurotransmitters?** Acetylcholine; monoamines (dopamine, norepinephrine, serotonin); amino acids (glutamate, GABA); peptides (endorphins); endocannabinoids.
13. **Difference between "physical" and "psychological" dependency?** Physical: an abstinence syndrome (opposite of the drug's effects) appears when the drug is abruptly stopped; it's a retrospective observation. Psychological: an ill-defined term for intense drug seeking from a keen desire for the drug (cannabis is the example). The two overlap more than the labels suggest.
14. **Define "set" and "setting."** Set: the user's expectations. Setting: the immediate environment where the drug is taken.

Chapter 3: Tolerance, dependence and behavior

1. **Define "tolerance."** A compensatory response of the whole system to the changes a drug causes, so the drug has less effect with repeated use and more is needed for the same effect. The body is seeking equilibrium.
2. **Define "metabolic tolerance."** The liver produces more enzymes, so the drug is broken down and cleared faster.
3. **Define "physiological" tolerance.** Same idea as pharmacodynamic tolerance: the nervous system adapts at the cellular level, e.g., changing the number or sensitivity of receptors at the synapse.
4. **Define "behavioral" tolerance.** Practicing a behavior while on a drug makes you better at it while on the drug. Treadmill rats: practicing while drunk improved drunk running; alcohol given after practice didn't.
5. **Define "learned" tolerance.** Compensatory responses triggered by environmental cues associated with taking the drug. Morphine room vs. saline room: the same morphine dose caused much less sedation in the familiar drug room. Chemo patients react to treatment-room cues.
6. **What is the "abstinence syndrome"?** The medical term for withdrawal: a constellation of responses opposite to the drug's effects when chronic use stops (e.g., alcohol, a sedative, causes overactivation in withdrawal).
7. **Difference between "physical" and "psychological" dependence?** See Chapter 2, question 13.
8. **Define "set" and "setting."** See Chapter 2, question 14. Stimulant example: focus in class vs. fun at a party.

Chapter 4: The placebo/expectation effect

1. **According to Andrew Weil, what's the difference between "set" and "setting"?** Set is the person's expectations and mindset going in; setting is the physical and social environment where the drug is taken. Weil argued these shape a drug's effects as much as the drug itself.

2. **Define "placebo."** Traditionally, a substance with no pharmacological effect given to placate a patient, or a medicine given for psychological reasons. Dr. Baker's view: placebo effects are robust, so a placebo isn't truly inert; it works through expectation and can be used ethically.
3. **Characterize "voluntary" behavior. (book)** Behavior consciously chosen and initiated, controlled by the frontal cortex. The chapter's point is that this same conscious, frontal control can alter other brain circuits, which is how expectation produces placebo effects.
4. **Describe one early demonstration of the placebo effect. "Surgery as Placebo" (1950s):** sham heart surgery for angina worked as well as real surgery, because patients' expectation of success drove recovery. (Beecher's WWII observations and "The Powerful Placebo," 1955, are other early examples in the book's references.)
5. **Contribution of the arthroscopic knee surgery study? (book)** Patients who got sham knee surgery (incisions only) improved as much as those who had the real procedure, up to two years later. It showed that even a surgical procedure's benefit can be largely placebo.
6. **Two developments that improved understanding of the placebo effect? (book)** (1) Double-blind, placebo-controlled trials, which separate drug effects from expectation; (2) brain imaging (PET, fMRI), which shows placebos change real brain activity and chemistry.
7. **Two brain systems studied for placebo mechanisms?** The reinforcement (reward) system (VTA, nucleus accumbens) and the pain-modulation system. The frontal cortex connects to both.
8. **Three principal brain areas of the midbrain reward system?** Ventral tegmental area (VTA), nucleus accumbens, and frontal cortex.
9. **Besides the frontal cortex and VTA, two other inputs to the nucleus accumbens?** Hippocampus and amygdala (also thalamus).
10. **Possible role of dopamine in the nucleus accumbens?** Signaling expected reward: motivation, "wanting" and craving. Expecting relief can release dopamine there.
11. **Significance of frontal cortex involvement in the placebo circuit?** The frontal cortex handles expectation and conscious thought, and it projects to the reward and pain circuits. So beliefs can physically change brain activity: it's "happening inside your head," but that doesn't mean it isn't real.
12. **Two neurotransmitter systems integrated in the nucleus accumbens?** Dopamine and glutamate (GABA and endogenous opioids are also involved).
13. **Besides the frontal cortex, two other cortical areas in the placebo effect? (book)** Anterior cingulate cortex and insula (orbitofrontal cortex is also often cited).
14. **Suspected cause of Parkinson's disease? (book)** Loss of dopamine neurons in the substantia nigra, which starves the nigrostriatal pathway (basal ganglia) of dopamine.
15. **Deep brain stimulation of what area relieves Parkinson's? (book)** The subthalamic nucleus (the globus pallidus is another target).
16. **Increased release of which transmitter enhances relief of Parkinson's symptoms?** Dopamine. Placebos given to Parkinson's patients have been shown to release dopamine in the striatum.
17. **How would you explain placebo takers having changed brain patterns?** Expectation, generated in the frontal cortex, drives real changes in the reward and pain circuits, such as dopamine and endogenous opioid release. The mind's activity is brain activity, so belief produces measurable physiological change.

18. **One limitation of relying only on cognitive therapy to change brain circuitry? (book)** It's slow and effortful, depends on the person's ability and motivation, and may not be strong enough alone for severe conditions; drugs can produce faster or larger changes.
19. **Define the "nocebo" effect. (book)** Negative expectations causing real harmful effects or side effects, e.g., people who expect side effects report more of them, even on a placebo.
20. **How does Gabapentin (Neurontin) relate to the ethics of placebo? (book)** Gabapentin has been widely prescribed off-label for conditions where trials show it barely beats placebo. That raises the question: if much of a drug's benefit is placebo, is prescribing it (with its side effects and cost) more ethical than an honest placebo?
21. **What issue does the Balanced Placebo design address? (book)** It separates a drug's pharmacological effect from the effect of expecting the drug. Subjects are told they're getting the drug or a placebo, and actually receive the drug or a placebo, in all four combinations (2×2).

Chapter 5: Drug classification

1. **Five types of sedative/hypnotic (CNS depressants)?** The book's table lists: barbiturates, benzodiazepines, other non-barbiturate sedatives, ethanol (alcohol), cannabis, and antihistamines. (The question says five; know the first five, and add antihistamines if you have room.)
2. **An example of each:** - Barbiturates: phenobarbital, secobarbital - Benzodiazepines: Valium (diazepam), Librium, triazolam, alprazolam - Other non-barbiturate sedatives: methaqualone, meprobamate - Ethanol: alcohol - Cannabis: marijuana, hashish - Antihistamines
3. **One naturally occurring opiate?** Morphine (or codeine).
4. **Difference between the two types of synthetic opiates? (book)** Semi-synthetic opiates are made by chemically modifying natural opiates (heroin is made from morphine). Fully synthetic opiates are built entirely in the lab with no natural opiate starting material (methadone, Demerol).
5. **Example of each synthetic type?** Semi-synthetic: heroin. Fully synthetic: methadone or Demerol (meperidine).
6. **Five types of stimulants?** Amphetamines (including methamphetamine); cocaine (including synthetic cocaine); methylxanthines; drugs used to treat hyperactivity; over-the-counter appetite suppressants.
7. **Examples of three stimulant types:** methylxanthines: caffeine, theophylline, theobromine; hyperactivity drugs: Ritalin, pemoline; appetite suppressants: phenylpropanolamine; amphetamine: methamphetamine.
8. **Two main classes of drugs for mental disorders?** Antidepressants and antipsychotics.
9. **Two subtypes of each:** Antidepressants: tricyclics, newer heterocyclics, MAO inhibitors. Antipsychotics: phenothiazines, butyrophenones.
10. **A specific example for each subtype:**
 - Tricyclics: imipramine, amitriptyline, desipramine
 - Heterocyclics: fluoxetine (Prozac), trazodone
 - MAO inhibitors: tranylcypromine (Parnate)
 - Phenothiazines: chlorpromazine (Thorazine), thioridazine (Mellaril)
 - Butyrophenones: haloperidol (Haldol)

11. **Differences between the types of hallucinogens?** They're grouped by chemical structure: those structurally similar to serotonin; those structurally similar to norepinephrine; and other agents that resemble neither.
12. **Examples of each:** serotonin-like: LSD, psilocin; norepinephrine-like: mescaline, MDMA ("ecstasy"); other: ibotenic acid, phencyclidine (PCP, "angel dust").
13. **Types of drugs used for anesthesia?** Gaseous (nitrous oxide); volatile (ether, halothane); parenterally administered (barbiturates, ketamine).
14. **Types of drugs used to treat epilepsy?** Hydantoins (diphenylhydantoin, Dilantin); barbiturates (phenobarbital); carbamazepine; valproic acid.
15. **Criteria for Schedule I?** High potential for abuse, no currently accepted medical use in the US, and a lack of accepted safety for use under medical supervision.
16. **Criteria for Schedule II? (book)** High potential for abuse, but a currently accepted medical use (often with severe restrictions); abuse may lead to severe psychological or physical dependence.
17. **Examples of Schedule I and II drugs? (book)** Schedule I: heroin, LSD, cannabis (federally), MDMA, psilocybin. Schedule II: cocaine, methamphetamine, morphine, fentanyl, oxycodone, methylphenidate (Ritalin), amphetamine (Adderall).
18. **Difference in criteria between Schedule III and IV?** Schedule III: abuse potential less than I and II; moderate to low physical dependence or high psychological dependence (e.g., ketamine, anabolic steroids, codeine combinations). Schedule IV: low abuse potential relative to III and limited dependence (e.g., benzodiazepines like Xanax and Valium).

Drug classification table (Textbook Table 5-1)

Class	Subtypes and examples
1. Sedatives and hypnotics (CNS depressants)	Barbiturates (phenobarbital, secobarbital); benzodiazepines (Valium, Librium, triazolam, alprazolam); other non-barbiturates (methaqualone, meprobamate); ethanol; cannabis; antihistamines
2. Opiates	Natural (morphine, codeine); synthetic (heroin, methadone, Demerol)
3. Stimulants	Amphetamine (incl. methamphetamine); cocaine; methylxanthines (caffeine, theophylline, theobromine); hyperactivity drugs (Ritalin, pemoline); OTC appetite suppressants (phenylpropanolamine)
4. Drugs for mental disorders	Tricyclic antidepressants (imipramine, amitriptyline, desipramine); heterocyclic antidepressants (fluoxetine/Prozac, trazodone); MAO inhibitors (tranylcypromine/Parnate); phenothiazines (chlorpromazine/Thorazine, thioridazine/Mellaril); butyrophenones (haloperidol/Haldol)
5. Hallucinogens	Serotonin-like (LSD, psilocin); norepinephrine-like (mescaline, MDMA); other (ibotenic acid, PCP)
6. Anesthetics	Gaseous (nitrous oxide); volatile (ether, halothane); parenteral (barbiturates, ketamine)
7. Anti-epilepsy drugs	Hydantoins (Dilantin); barbiturates (phenobarbital); carbamazepine; valproic acid

Memory hook for the seven classes: **"Sleepy Owls Stay Mentally Hazy And Excited"**: Sedatives, Opiates, Stimulants, Mental-disorder drugs, Hallucinogens, Anesthetics, Epilepsy drugs.